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Early outcomes after acute type A aortic dissection repair: Adequate perfusion, rather than cannulation strategy

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ABSTRACT

Background: The optimal arterial cannulation strategy for acute type A aortic dissection repair remains uncertain. We aimed to evaluate the association between cannulation strategy and early postoperative outcomes and to identify perioperative factors associated with adverse events.

Methods: A single-center, retrospective cohort study was conducted at West China Hospital, Sichuan University, between January 2019 and June 2024, including consecutive patients undergoing repair of acute type A aortic dissection. Primary outcomes were in-hospital death and postoperative neurologic complications. Multivariable logistic regression and propensity-score matching were used to evaluate associations with early outcomes.

Results: Among 866 patients, femoral, axillary, and combined femoral + axillary cannulation were used in 694 (80.1%), 144 (16.6%), and 28 (3.2%) patients, respectively. Patients in the axillary and combined groups more frequently had primary arch tears and underwent more extensive arch repair. In-hospital death occurred in 43 (5.0%) patients and postoperative neurologic complications in 113 (13.0%) patients, with comparable rates across groups; paraplegia was more frequent in the combined group (2/28 [7.1%]; $P = .03$). In multivariable analysis, axillary cannulation was not independently associated with in-hospital death (odds ratio, 1.74; 95% confidence interval, 0.69–4.03), postoperative neurologic complications (odds ratio, 0.73; 95% confidence interval, 0.37–1.34), or cerebral infarction (odds ratio, 1.05; 95% confidence interval, 0.51–2.01). Preoperative shock (odds ratio, 3.20; 95% confidence interval, 1.44–6.91) and higher peak intraoperative lactate (odds ratio, 1.22; 95% confidence interval, 1.14–1.30) were independently associated with in-hospital death. In 142 propensity score–matched femoral-axillary pairs, mortality and neurologic complication rates remained similar.

Conclusion: Cannulation strategy was not independently associated with in-hospital death or postoperative neurologic complications. Preoperative shock and higher peak intraoperative lactate were more consistently associated with adverse early outcomes, supporting the clinical importance of timely systemic reperfusion and effective cerebral protection.

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Introduction

Acute type A aortic dissection (ATAAD) is a life-threatening condition that requires urgent surgical intervention.[1] Despite substantial advances in surgical techniques and perioperative management over recent decades, in-hospital mortality and major complications remain unacceptably high, imposing a significant burden on affected patients. Cerebral infarction is the most common postoperative complication following ATAAD repair and represents a leading cause of early mortality. The occurrence of neurologic injury is multifactorial but is closely related to the extent of aortic dissection, involvement of the supra-aortic vessels, and the complexity of aortic arch procedures.[2–4]

Arterial cannulation strategy is a fundamental technical decision in ATAAD surgery and includes femoral, axillary, central aortic,[5] and carotid artery[6] cannulation, with selection largely determined by surgeon preference and institutional practice. Femoral and axillary

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arteries are the most commonly used sites, each with theoretical advantages and potential drawbacks.[7–10] Axillary cannulation offers antegrade systemic flow and readily supports antegrade cerebral perfusion (ACP), whereas femoral cannulation is fast, technically straightforward, and often favored in hemodynamically unstable patients. Use of axillary cannulation has increased over recent decades,[9] driven by the perception that the axillary artery is less often compromised by the dissection and by reports suggesting lower stroke rates. However, comparative data remain mixed, and cannulation choice is frequently guided by surgeon preference, institutional practice, and patient anatomy rather than definitive evidence. Many prior comparisons also reflect earlier operative eras and heterogeneous techniques. Therefore, contemporary data from high-volume centers remain relevant for reassessing the association between cannulation site and early outcomes.

This study analyzed outcomes in a large, single-center ATAAD cohort to clarify whether axillary or femoral cannulation was associated with differences in mortality or neurologic outcomes after adjusting for patient and operative complexity. We also evaluated whether markers of hypoperfusion burden, including preoperative shock and peak intraoperative lactate, were associated with adverse early outcomes.

Methods

This single-center retrospective study conformed to the ethical guidelines of the 1975 Declaration of Helsinki and was approved by the Biomedical Ethics Committee of West China Hospital, Sichuan University (no. 2025 Review [1706], approved on August 16, 2025). Patient consent for the publication of the study data was waived by the institutional review board because of minimal risk to patients. This retrospective cohort study is reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement. A completed STROBE checklist is provided in Supplementary File 1.

Study population and data collection

The present study was based on the West China Hospital Open Aortic Repair for Dissection (WARD) cohort, which prospectively enrolled consecutive patients undergoing surgery for aortic dissection from 2019 onward.[11] For the present analysis, we reviewed consecutive patients who underwent open repair for ATAAD between January 2019 and June 2024. ATAAD was diagnosed by computed tomography angiography and confirmed by operative findings. Patients with isolated intramural hematoma, penetrating aortic ulcer, previous aortic surgery,[12] type B dissection, chronic type A dissection, or other cannulation strategies were excluded. No age restrictions were applied. Patients were classified according to arterial cannulation strategy as femoral, axillary, or combined femoral + axillary cannulation. Femoral arterial cannulation was established using the Seldinger technique. Axillary cannulation was performed without a prosthetic side graft (direct cannulation) either with or without the Seldinger technique. The combined femoral plus axillary group included patients in whom both access sites were used, either as a planned strategy or because of intraoperative conversion or addition of a second arterial access site. The final cohort consisted of 866 patients, and the full study flow diagram is shown in Figure 1.

Outcomes

Primary outcomes were in-hospital mortality and postoperative neurologic complications (PNCs). PNCs were defined as any new neurologic event occurring within the perioperative period, including cerebral infarction, intracerebral hemorrhage, paraplegia, hypoxic-ischemic encephalopathy, or cerebral edema. Cerebral infarction was defined as a new focal neurologic deficit with corresponding ischemic lesion on computed tomography (CT) or magnetic resonance imaging (MRI), or a neurologist-documented diagnosis when imaging was not feasible. Intracerebral hemorrhage was defined as new postoperative intracranial bleeding confirmed by CT or MRI. Paraplegia was defined as new postoperative bilateral lower-extremity motor dysfunction consistent with spinal cord injury and not attributable to cerebral stroke. Hypoxic-ischemic encephalopathy was defined as new postoperative global neurologic dysfunction or coma attributed to diffuse hypoxic-ischemic brain injury. Cerebral edema was defined as new postoperative diffuse or regional brain swelling on CT or MRI reported by a radiologist or a neurologist.

Secondary outcomes were peak intraoperative lactate, nadir temperature, operative duration, cardiopulmonary bypass (CPB) time, aortic cross-clamp time, and circulatory arrest time. Perioperative outcomes were defined as events occurring between surgery and discharge. Cannulation-related access complications were additionally reviewed across the entire cohort. These were defined as cannulation-site bleeding requiring surgical revision or re-exploration, iatrogenic arterial injury or dissection, pseudoaneurysm, limb ischemia, cannulation failure requiring conversion or additional arterial access, brachial plexus injury, and clinically relevant wound complications.

Statistical analyses

All analyses were performed using R software, version 4.4.3 (R Foundation for Statistical Computing, Vienna, Austria). Continuous variables were tested for normality with the Shapiro-Wilk test and are presented as mean (standard deviation) or median (interquartile range), as appropriate. Between-group comparisons used 1-way analysis of variance or the Kruskal-Wallis test for continuous variables, and the χ^2 test or the Fisher exact test for categorical variables. Propensity scores for axillary cannulation were estimated using logistic regression with prespecified baseline and anatomic covariates, followed by 1:1 nearest neighbor matching with a caliper of 0.1 standard deviations of the logit of the propensity score. Balance after matching was assessed using standardized mean differences. In the matched cohort, paired comparisons used paired *t* tests or Wilcoxon signed-rank tests for continuous variables and McNemar tests for binary outcomes. Multivariable logistic regression was used to evaluate the association between cannulation strategy and postoperative outcomes, with femoral cannulation as the reference. Covariates were selected on the basis of clinical relevance, baseline imbalance, and model parsimony; Firth penalized logistic regression was used when event counts were low or separation was present. Results are reported as odds ratios with 95% confidence intervals. All tests were 2 sided.

Results

Baseline characteristics

Between January 2019 and June 2024, 866 patients who underwent open ATAAD repair were included. Femoral cannulation remained the predominant arterial access strategy ($n = 694$ [80.1%]). The others had axillary ($n = 144$ [16.6%]) and combined femoral plus axillary ($n = 28$ [3.2%]) cannulation (Supplementary Figure). Baseline demographic characteristics were similar across groups. Valvular heart disease and alcohol intake were more frequent in the combined cannulation group. The distribution of primary tear site differed by cannulation strategy: ascending aortic tears were most common overall but less frequent in the axillary

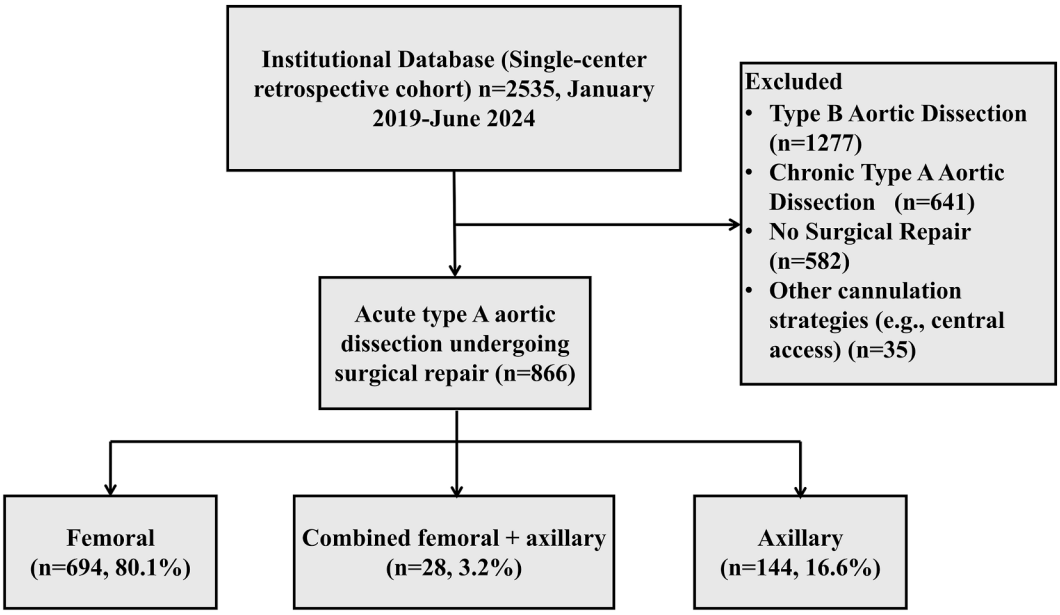


Figure 1. Flowchart of patient selection and grouping (2019.1–2024.6). *ATAAD*, acute type A aortic dissection.

and combined groups, whereas arch tears were more prevalent in these groups. Malperfusion at presentation differed among groups; among malperfusion subtypes, only extremity malperfusion differed significantly, whereas superior mesenteric and neurologic malperfusion were similar across groups. Baseline characteristics are summarized in Table I.

Operative characteristics

Operative extent differed substantially according to cannulation strategy: Total arch replacement was more common with axillary and combined cannulation, whereas hemiarch replacement was performed most frequently in the femoral group. Aortic valve repair was more common in the axillary group ($P = .02$), whereas rates of aortic valve replacement were similar across groups. Concomitant coronary artery bypass grafting, ligation of the left subclavian artery, and bovine arch anatomy also did not differ significantly. Operative duration showed a borderline difference and tended to be longest in the combined group ($P = .05$). Peak intraoperative lactate levels differed across strategies (femoral 6.7 ± 4.1, axillary 5.8 ± 3.5, combined 7.8 ± 4.6; $P = .03$), and circulatory arrest time also varied significantly ($P < .001$), whereas CPB time, cross-clamp time, lowest intraoperative temperature, and left ventricular ejection fraction were comparable. Cerebral protection strategies differed markedly by cannulation approach ($P < .001$): femoral cannulation more frequently involved no cerebral perfusion, whereas axillary and combined cannulation predominantly used unilateral ACP; retrograde cerebral perfusion (RCP) was uncommon in the axillary and combined groups. Operative characteristics are summarized in Table II.

Cannulation-related access complications were reviewed; access-related complications occurred in 7 patients in the femoral group, 1 in the axillary group, and 0 in the combined group. Specific events are reported separately in Supplementary Table S1.

Early outcomes

In the unadjusted comparison, in-hospital mortality was low and did not differ by cannulation strategy (5.0% vs 5.6% vs 0%; $P = .59$). Overall PNCs were also similar across groups (13.8% vs 9.0% vs 14.3%; $P = .27$), driven largely by comparable rates of cerebral infarction ($P = .97$) and intracerebral hemorrhage ($P = .46$). Notably, paraplegia was uncommon overall but occurred more frequently in the combined group ($P = .03$). Paraplegia was uncommon overall but was observed more frequently in the combined femoral plus axillary group (1.3% vs 0% vs 7.1%; $P = .03$), although the absolute number of events was small. Other neurologic complications were infrequent and not significantly different (Supplementary Table S2).

After multivariable adjustment, cannulation strategy was not independently associated with in-hospital death (Figure 2). The estimate for combined femoral plus axillary cannulation was unstable because no deaths occurred in this subgroup and should therefore be interpreted cautiously. Instead, early mortality was independently associated with preoperative shock (odds ratio [OR], 3.20; confidence interval [CI], 1.44–6.91; $P < .001$) and higher peak intraoperative lactate levels (OR, 1.22; CI, 1.14–1.30; $P < .001$), with a borderline association observed for longer aortic cross-clamp time ($P = .05$). The estimate for the combined group was unstable because no deaths occurred in that subgroup ($P = .98$). For neurologic outcomes, neither axillary nor combined cannulation was associated with overall PNCs or cerebral infarction. Instead, neurologic outcomes were more consistently associated with patient and operative factors: older age (OR, 1.02 per year; CI, 1.00–1.04; $P = .02$), ligation of the left subclavian artery (OR, 2.42; CI, 1.04–5.23; $P = .03$), and higher peak intraoperative lactate levels (OR, 1.13; CI, 1.08–1.18; $P < .001$). Higher peak intraoperative lactate levels were independently associated with cerebral infarction (OR, 1.1; $P < .001$), intracerebral hemorrhage (OR, 1.11; $P = .04$), and paraplegia (OR, 1.14; $P = .04$) (Supplementary Table S3).

Because cerebral perfusion strategy differed substantially by cannulation site, we performed sensitivity analyses for PNCs and cerebral infarction with additional adjustment for cerebral perfusion strategy. In these analyses, the association between cannulation strategy and neurologic outcomes remained unchanged. Axillary cannulation remained not associated with PNCs (OR, 0.72; 95% CI, 0.36–1.33; $P = .32$) or cerebral infarction (OR, 0.99; 95% CI,

Table I
Baseline characteristics based on cannulation strategy

Variable	Femoral ($n = 694$)	Axillary ($n = 144$)	Femoral + axillary ($n = 28$)	*P*
Age, y	51.9 ± 12.1	51.8 ± 11.8	49.0 ± 11.9	.47
Female sex	179 (25.8)	30 (20.8)	3 (10.7)	.10
BMI, kg/m^2	25.2 ± 3.9	25.0 ± 3.8	25.2 ± 3.4	.75
Previous surgery	114 (16.4)	22 (15.3)	1 (3.6)	.16
Pre-existing comorbidities				
Hypertension	489 (70.5)	41 (28.5)	21 (75.0)	.85
Coronary artery disease	17 (2.4)	4 (2.8)	1 (3.6)	.71
History of cerebrovascular accident	27 (3.9)	2 (1.4)	1 (3.6)	.28
Thrombosis	3 (0.4)	0 (0)	0 (0)	1.00
Arrhythmia	14 (2.0)	6 (4.2)	1 (3.6)	.20
Renal dialysis	44 (6.3)	9 (6.3)	2 (7.1)	.96
Diabetes mellitus	20 (2.9)	5 (3.5)	1 (3.6)	.68
Carotid atherosclerosis	5 (0.7)	2 (1.4)	0 (0)	.48
Chronic obstructive pulmonary disease	30 (4.3)	7 (4.9)	3 (10.7)	.22
Valvular heart disease	80 (11.5)	18 (12.5)	10 (35.7)	<.001
Thyroid disease	14 (2.0)	0 (0)	1 (3.6)	.15
Smoking history	268 (38.6)	55 (38.2)	15 (53.6)	.28
Alcohol intake history	201 (29.0)	49 (34.0)	14 (50.0)	.04
Marfan syndrome	17 (2.4)	6 (4.2)	1 (3.6)	.35
Rupture/tamponade	37 (5.3)	3 (2.1)	1 (3.6)	.23
Shock	90 (13.0)	19 (13.2)	6 (21.4)	.42
Primary tear site				<.001
Ascending aorta	517 (74.5)	87 (60.4)	15 (53.6)	
Arch	134 (19.3)	46 (31.9)	13 (46.4)	
Descending aorta	25 (3.6)	10 (6.9)	0 (0)	
Any malperfusion	207 (29.8)	36 (25.0)	18 (64.3)	.02
Superior mesenteric	24 (3.5)	7 (4.9)	2 (7.1)	.56
Extremity	107 (15.4)	16 (11.1)	10 (35.7)	.01
Neurologic	76 (11.0)	13 (9.0)	6 (21.4)	.16

Data are presented as mean ± standard deviation for continuous variables and *N* (%) for categorical variables. N indicates the number of positive cases, and % indicates the percentage of positive cases over the whole cohort ($n = 866$).
BMI, body mass index.

Table II
Intraoperative characteristics based on cannulation strategy

Variable	Femoral ($n = 694$)	Axillary ($n = 144$)	Femoral + axillary ($n = 28$)	*P*
Surgical procedure				
Bentall	283 (40.8)	46 (31.9)	13 (46.4)	.11
Root preserved	315 (45.4)	84 (58.3)	14 (50.0)	.02
Total arch replacement	455 (65.6)	125 (86.8)	27 (96.4)	<.001
Hemi arch replacement	170 (24.5)	16 (11.1)	1 (3.6)	<.001
Coronary artery bypass grafting	58 (8.4)	16 (11.1)	4 (14.3)	.32
Ligation of the left subclavian artery	35 (5.0)	3 (2.1)	2 (7.1)	.19
Bovine aortic arch	8 (1.2)	4 (2.8)	0 (0)	.26
Left ventricular ejection fraction	62.2 ± 9.4	63.3 ± 9.1	62.5 ± 10.0	.42
Peak intraoperative lactate level	6.7 ± 4.1	5.8 ± 3.5	7.8 ± 4.6	.03
Lowest intraoperative temperature	24.9 ± 3.1	24.0 ± 1.2	24.2 ± 1.0	.86
Operative duration	425.9 ± 94.8	428.2 ± 80.8	454.2 ± 60.2	.05
CPB time	202.9 ± 58.7	203.3 ± 52.7	207.3 ± 45.7	.65
Aortic cross-clamp time	148.9 ± 49.8	147.2 ± 47.8	153.7 ± 34.5	.46
Circulatory arrest time	24.8 ± 13.5	22.7 ± 8.8	27.1 ± 14.4	<.001
Cerebral perfusion strategy				<.001
No cerebral perfusion	114 (16.4)	5 (3.5)	0 (0)	
ACP (unilateral)	383 (55.2)	121 (84.0)	27 (96.4)	
ACP (bilateral)	112 (16.1)	14 (9.7)	0 (0)	
RCP	3 (12.2)	4 (2.8)	1 (3.6)	

Data are presented as mean ± standard deviation for continuous variables and *N* (%) for categorical variables. *N* indicates the number of positive cases, and % indicates the percentage of positive cases over the whole cohort ($n = 866$).
ACP, antegrade cerebral perfusion; *CPB*, cardiopulmonary bypass.

0.50–1.97; $P = .99$). Combined femoral plus axillary cannulation also remained nonsignificant for PNCs (OR, 0.87; 95% CI, 0.24–2.48; $P = .82$) and cerebral infarction (OR, 0.58; 95% CI, 0.13–2.62; $P = .48$). These findings suggest that the main neurologic results were robust to additional adjustment for cerebral perfusion strategy (Supplementary Table S4).

Propensity-matched comparison

In the propensity score–matched cohort (142 pairs), early outcomes were comparable between groups (Figure 3). Mortality was similar (4.2% vs 5.6%; $P = .77$), and there were no significant differences in overall PNCs (14.1% vs 9.2%; $P = .16$) or specific

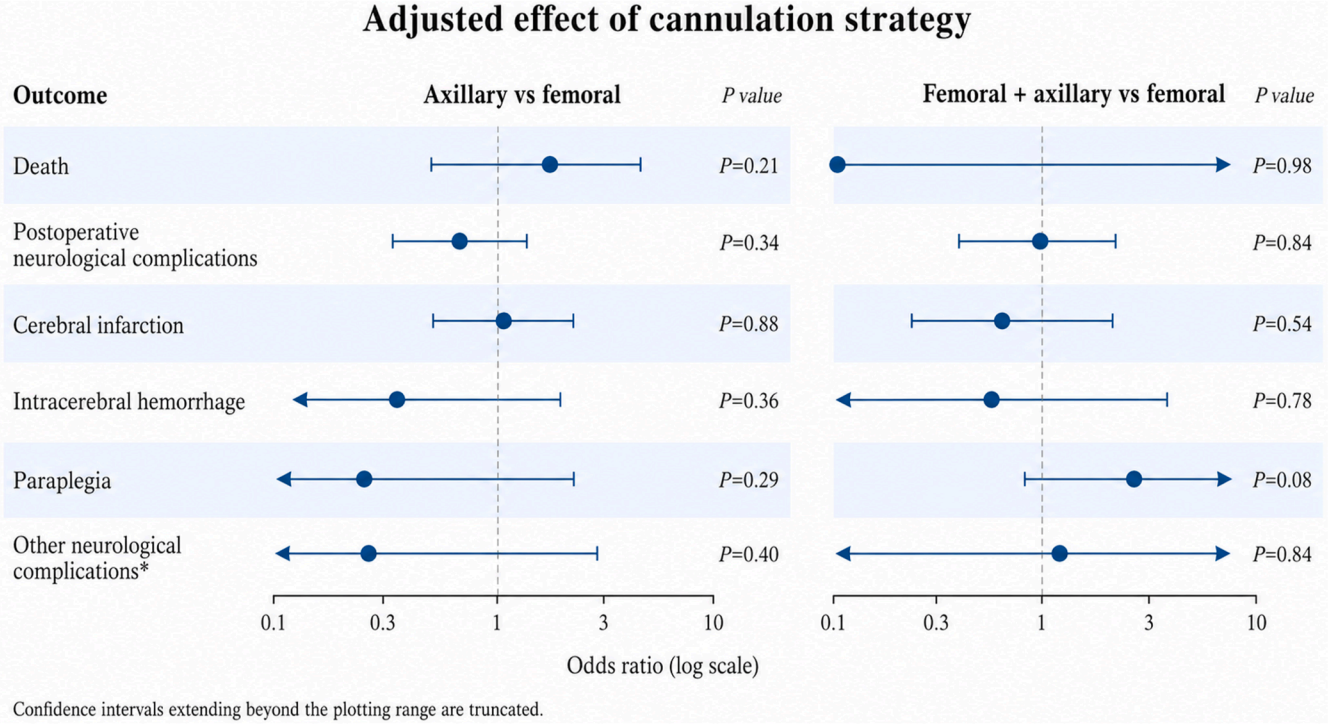


Figure 2. Adjusted associations between cannulation strategy and early postoperative outcomes after acute type A aortic dissection repair.

neurologic events. Among matched patients, axillary cannulation was associated with lower peak intraoperative lactate and shorter circulatory arrest time, whereas operative duration, CPB time, aortic cross-clamp time, and nadir temperature were similar. Baseline covariate balance and postmatching intraoperative characteristics are summarized in Supplementary Table S5.

Discussion

In this large single-center ATAAD cohort, arterial cannulation strategy was not independently associated with early mortality or overall PNCs. This finding was consistent across unadjusted analyses, multivariable regression, and propensity score matching, suggesting that cannulation site alone was not independently associated with early outcome when contemporary perfusion and cerebral protection strategies are applied. In contrast, preoperative shock and peak intraoperative lactate were more consistently associated with adverse early outcomes, highlighting the prognostic relevance of hypoperfusion burden.

Femoral cannulation was first described in 1959 and became widely adopted in aortic surgery.[13] However, concerns remain that retrograde perfusion may increase the risk of false-lumen flow, malperfusion, and embolic stroke. Right axillary cannulation was introduced to address these limitations in the 1990s.[14] With the widespread adoption of selective ACP around the early 2000s, axillary cannulation gained increasing popularity.[7–10,15,16] Nevertheless, existing comparative evidence has largely been derived from small, single-center studies or large, registry-based analyses and remains heterogeneous. As a result, the relative merits of these 2 strategies remain controversial. Despite evolving practice, femoral cannulation continues to be commonly used in ATAAD surgery at most centers. Accordingly, access-site selection must be interpreted in the context of patient physiology, operative extent, cerebral protection, and institutional workflow.

Femoral cannulation remained the dominant strategy in our practice, likely reflecting ease and speed of access, patient body habitus, anatomic suitability, surgeon preference, and institutional practice patterns. Patients in our cohort were relatively young and had moderate body mass index, which may have facilitated rapid femoral artery exposure.[17–20] In such settings, Seldinger-guided femoral cannulation allowed prompt establishment of CPB before extensive manipulation of the fragile proximal aorta.[15] The use of direct Seldinger-guided femoral cannulation enabled reliable delivery of CPB flow while minimizing the risks of vasospasm, small-caliber vessel limitation, or excessive pump pressures, thereby facilitating timely reperfusion in hemodynamically unstable patients.[21,22]

In our practice, axillary cannulation is reserved for selected anatomic and perfusion-related scenarios rather than used routinely. These include cases in which both femoral arteries are involved by dissection; the femoral artery appears intact, but the proximal perfusion is supplied predominantly by the false lumen[23]; and patients with large intimal tears involving the iliac arteries or distal abdominal aorta.[24,25] In addition, axillary access is favored in patients with small-caliber femoral arteries or significant peripheral arterial disease.[14,26,27] In selected cases with extensive distal disease, axillary cannulation is combined with central or innominate artery cannulation to ensure adequate antegrade systemic and cerebral perfusion.[22,25,28]

Of note, Svensson et al[29] cautioned that direct cannulation may increase the risk of difficult-to-repair arterial injury and may provide inadequate CPB flow in some patients. Nevertheless, despite greater anatomic complexity and operative extent in our cohort, early outcomes were comparable to those reported in prior studies, suggesting that direct axillary cannulation can be feasible in selected patients in an experienced center. This may partly relate to our routine use of larger arterial cannulas (>18 Fr, most commonly 19–21 Fr), which facilitates adequate CPB flow delivery and reduces flow limitation in selected patients.[30] However, because this study was not designed to compare direct axillary

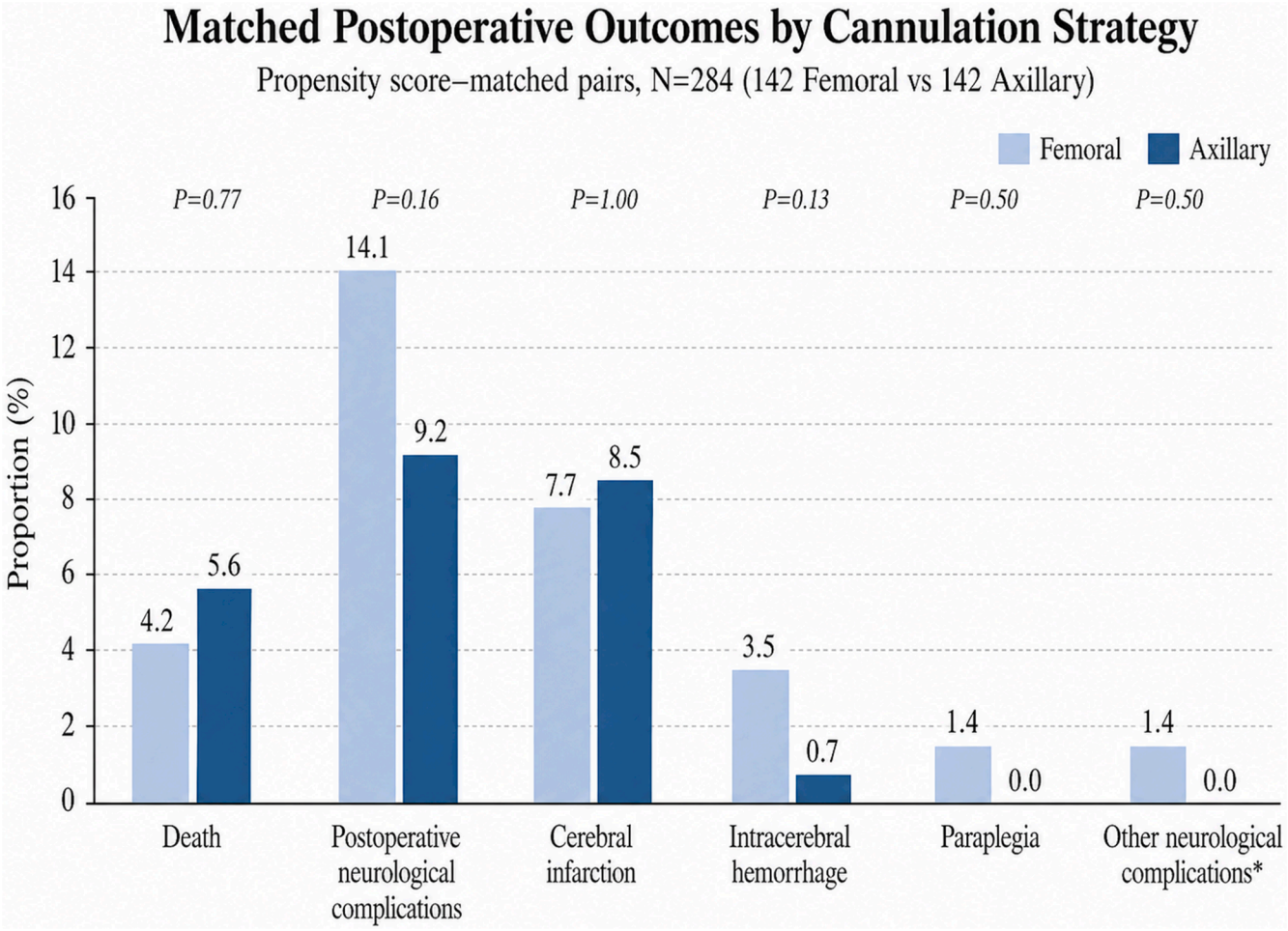


Figure 3. Propensity score–matched comparison of femoral versus axillary cannulation after acute type A aortic dissection repair.

cannulation with side-graft cannulation, no conclusion can be drawn regarding the relative safety of these 2 axillary techniques.

These cases were also associated with distinct cerebral perfusion strategies. Axillary cannulation was more often combined with ACP, whereas femoral cannulation was more frequently associated with either no cerebral perfusion or RCP. In our practice, femoral cannulation is more commonly selected when a hemiarch repair is anticipated, and RCP is preferentially used in these cases. Koprivanac et al[31] suggested that axillary cannulation with unilateral ACP may mitigate cerebral injury risk by avoiding direct balloon-catheter perfusion of the head vessels when backflow is limited, a setting in which manipulation can introduce air or dislodge atheromatous debris. We applied direct cannulation through the innominate artery and/or left common carotid artery for ACP with a perfusion flow rate of 6–12 mL min^{-1} kg^{-1}, which allows rapid, stable cerebral flow support during circulatory arrest and facilitates timely flexible adjustment based on real-time monitoring.[32,33] Despite these differences, CPB time and aortic cross-clamp time were similar between groups, likely because the cohort was still dominated by total arch procedures with unilateral ACP. To address this potential confounding factor, we performed sensitivity analyses additionally adjusting for cerebral perfusion strategy. These analyses did not materially change the association between cannulation strategy and neurologic outcomes.

Preoperative shock and higher peak intraoperative lactate were the most consistent factors associated with early death and neurologic injury.[4,8,34,35] These findings emphasize the importance of rapid hemodynamic stabilization and timely correction of hypoperfusion. After propensity matching, patients in the axillary group exhibited lower peak lactate levels. Peak intraoperative lactate should not be interpreted as an isolated causal factor. It likely integrates preoperative shock burden, end-organ malperfusion, operative complexity, and the timeliness and adequacy of systemic reperfusion.[36] From a practical perspective, strategies that reduce the severity and duration of hypoperfusion may be more important than focusing solely on the arterial access site.

Current guidelines tend to favor axillary or central cannulation largely on the basis of retrospective studies reporting lower stroke risk.[7,25,27] However, these data remain vulnerable to confounding by indication, which limits causal interpretation. In major registries and meta-analyses, axillary cannulation is more often selected in stable patients and in experienced aortic centers, whereas femoral cannulation is frequently used when rapid bypass establishment is required.[25] Our findings support the principle that cannulation strategy should provide rapid and reliable CPB and effective cerebral perfusion but do not support the view that any single access site is inherently superior. The combined cannulation subgroup was small and included both planned use and intraoperative conversion, which could not be separated in this analysis. This limits interpretation of that subgroup. Notably, PNCs combine cerebral infarction, hemorrhage, paraplegia, hypoxic-ischemic encephalopathy, and cerebral edema; the

composite should be interpreted as a broad marker of neurologic morbidity rather than a uniform clinical entity. Functional status and neurocognitive outcomes were not available and represent important targets for future studies.

Overall, our results suggest that early outcome is associated more consistently with physiologic severity, perfusion adequacy, and cerebral protection than with cannulation site alone.

Limitations

This study has several limitations. First, the retrospective single-center design is vulnerable to selection bias, and residual confounding remains possible despite multivariable adjustment and propensity matching. Second, the combined subgroup was small, with few events and imprecise estimates. Third, cannulation selection was not random and likely reflected anatomy, hemodynamic urgency, malperfusion pattern, anticipated arch extent, and institutional workflow. Therefore, the observed associations should not be interpreted as causal effects of access site. Fourth, cannulation site was closely linked to operative strategy and cerebral protection, limiting the ability to isolate the association of any single intraoperative component. Fifth, PNCs represented a composite end point that included events with different mechanisms, severity, and functional implications. Sixth, only in-hospital outcomes were available; long-term survival, late neurologic events, neurocognitive status, and functional outcomes were not assessed. Nevertheless, these data provide contemporary real-world evidence and support further multicenter studies with more detailed procedural data and longer follow-up. Finally, sex-specific differences were not formally evaluated. Although sex was included as a baseline variable, the study was not designed or powered to assess sex- or gender-based interactions, and no stratified analyses were performed.

In conclusion, in this single-center cohort of ATAAD repair, cannulation strategy was not independently associated with early mortality or overall neurologic complications. Early adverse outcomes were more closely related to perioperative shock and higher peak intraoperative lactate, supporting the clinical importance of rapid restoration of systemic perfusion and effective cerebral protection. Cannulation strategy should be individualized according to patient anatomy, hemodynamic urgency, operative plan, and institutional expertise.

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Conflict of Interest/Disclosure

The authors have no conflicts of interest to declare.

Data availability

The data that support the study findings are available upon reasonable request from the corresponding authors.

CRediT authorship contribution statement

Xiangfeng Gong: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Julin Zhang:** Writing – original draft, Visualization, Formal analysis, Data curation, Conceptualization. **Wei Meng:** Visualization, Formal analysis. **Jia Hu:** Resources, Investigation. **Zhong Wu:** Resources, Investigation. **Zhenghua Xiao:** Writing – review & editing, Supervision, Project administration. **Chaoyi Qin:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [https://doi.org/10.1016/j.surg.2026.110519].

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